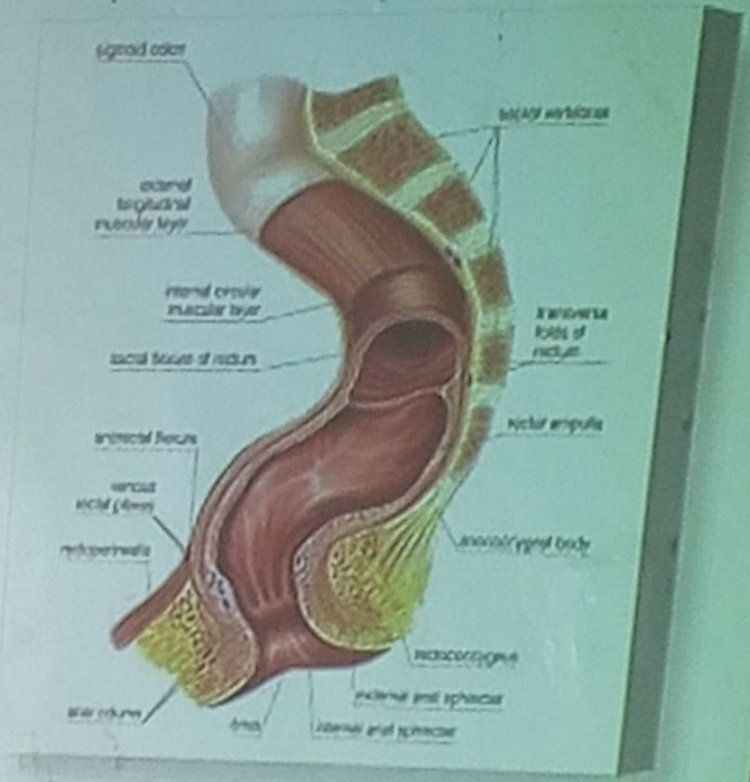
Rectum

- Introduction
- Shape and location
- Curves in the rectum
- Peritoneal reflection
- Mucous folds
- Relations of peritoneum in both sexes
- Blood supply
- Innervations
- Clinical anatomy

Activate Windows

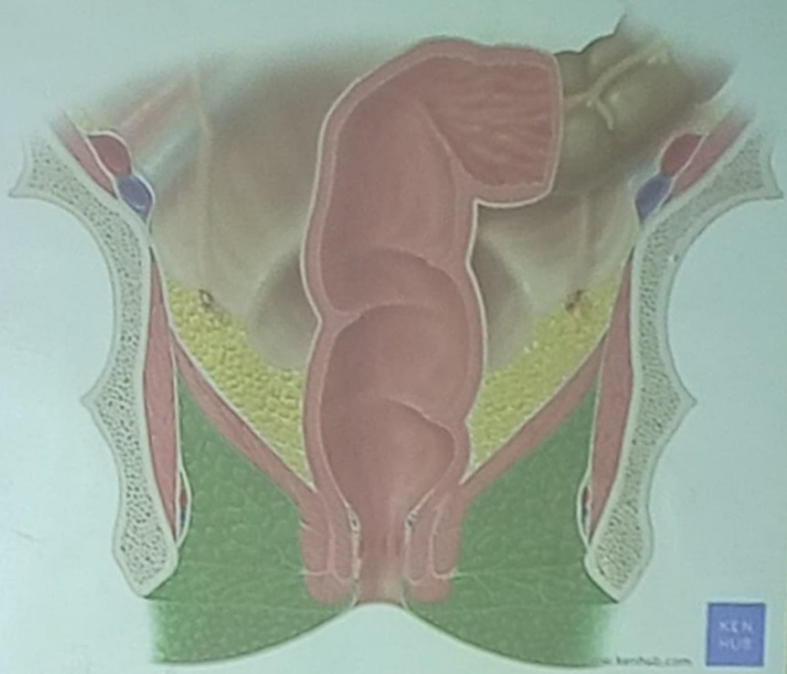
Rectum: Rectum means straight. Though not straight in adult human but are still straight in lower animals and children. It's length is 12cm and has a diameter of 4cm equal to sigmoid colon except at the ampulla part. It is a continuation of sigmoid colon from 3rd sacral segment to slightly below the tip of coccyx where it bends abruptly posterior continuing as anal canal. The beginning and the end of rectum lie in the median plane but shows two types of curvatures in its course.



Activate Windows

(a) Two anterior posterior curves: (1) Sacral flexure of rectum follow the concavity of sacrum and coccyx. (2) The perineal flexure of rectum is the backward bend at the anorectal junction.

(b) Three transverse or lateral curves: The upper and lower lateral curves of rectum are convex to the right while the middle lateral curves which is most prominent is convex to the left.

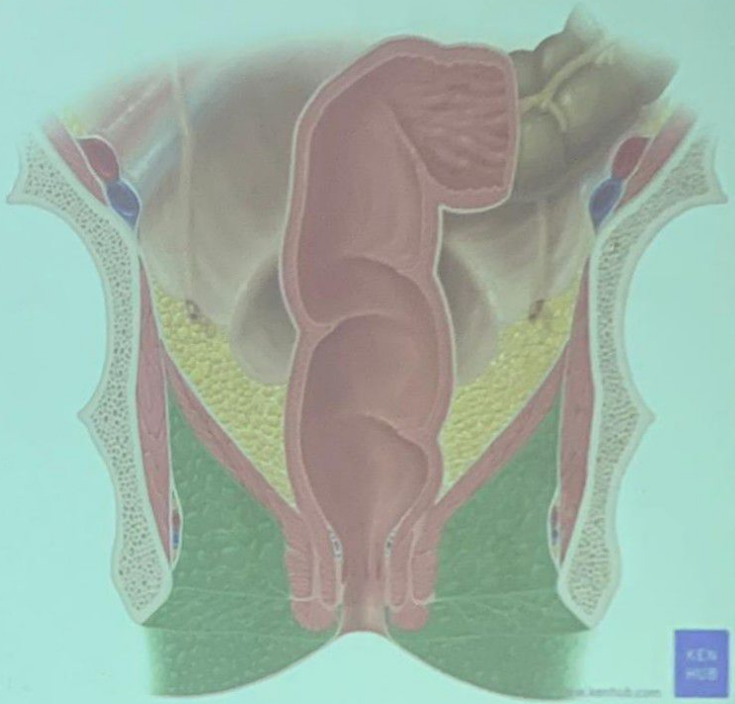


Activate Windows

(c) Peritoneal reflection of rectum: (1) The upper 1/3 is covered by peritoneum from the front and the sides. (2) The middle 1/3 is covered by peritoneum only in front. (3) The lower 1/3 has no peritoneal at all.

(d) Mucous folds: The interior of rectum presents two types of folds; temporal and permanent. The temporal fold are longitudinally placed folds of mucosa which disappear on the distension of the rectum. Horizontal folds are formed by mucous membrane and circular muscle coat of the rectum. They are three in number (valves of Houston's).

Upper fold lies near the upper end of rectum and projects from the right to the left wall or encircle the lumen.



Activate Windows

Middle fold which is the largest and most consistent lies at the upper end of the rectal ampulla and projects from the anterior and right wall.

Lower fold very inconsistent lies 2.5cm below the middle fold and projects from the left wall.

Relations of rectum in males and in females:

Males anterior; Urinary bladder, seminal vesicles, ureters, posterior surface of prostate, urethra and ampulla of vas deferens.

Female anterior: Pouch of Douglas containing coils of small intestine, , sigmoid colon, directly relates with the lower part of uterus, cervix and upper par of vagina. .

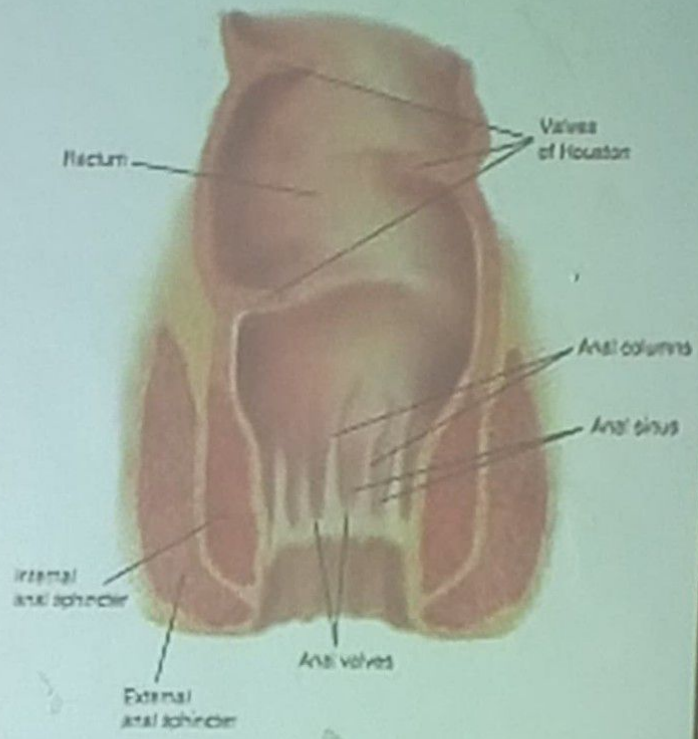


Figure 39-3 Anal Sphincter .

Activate Windows

Posterior relation: The relation of rectum is same for both sexes at this part.

Bones; lower 3 pieces of sacrum, coccyx, anococcygeal ligament.

Muscles; Piriformis, coccygeus and levator ani.

Vessels; Median sacral, superior rectal, and lower lateral sacral vessels.

Nerves; Sympathetic chain, ganglia impar, sacral nerves 3-5, coccygeal nerves, pelvic splanchnic nerve.

Fibrous tissue: Fibro-fatty tissue and lymph nodes.

Blood supply: Superior rectal artery, continuation of inferior mesenteric artery, middle rectal artery, branch of anterior division of internal iliac artery, inferior rectal artery, branch of internal pudendal artery.

Activate Windows

Median sacral artery, branch of aorta at its bifurcation.

Venous drainage: Haemorrhoidal veins drain upper part of anal canal above dentate line known as rectal vein. They join to form superior rectal vein which becomes inferior mesenteric vein and joins splenic vein which units with superior mesenteric vein to form portal vein. The middle rectal vein, small and becomes important only in the incidence of blockage.

Lymphatic drainage: (1) Para-rectal nodes of Gerota drains at the posterior of ampulla. (2) Middle rectal nodes drains into internal iliac nodes.

Nerve supply: It is supplied by both sympathetic (L1-L2), and parasympathetic (S2- S3 or S4) through inferior hypogastric plexus. The sympathetic nerve is vaso-constrictor and inhibitory to rectal musculature, parasympathetic is motor to muscles of rectum.

Activate Windows

Differences between rectum and sigmoid colon:

Sigmoid colon	Rectum
Sacculations	No sacculations
Appendices epiploicae	None
Mesentary	None
Taenia coli in three bands	Taenia coli in two bands
Peritoneum intimately attached to the organ.	Peritoneum loosely attached to the organ.

Clinical Anatomy: Sigmoid colon is the faecal reservoir and the whole of rectum is empty in normal individuals, being sensitive to distension. Passage of faecal content into the rectum causes the desire to defecate.

Activate Windows